

CO2 - AT3 -SCENARIO BASED QUESTONS

**Course Title: Machine Learning Course
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1. In the context of neural networks, consider a multilayer perceptron used for binary classification. Explain how the backpropagation algorithm updates the weights in the network during training. Illustrate the process using a simple example by describing the forward pass, error calculation, and backward propagation of errors. Discuss how learning rate and activation functions influence the convergence and accuracy of the model.?

1. Introduction :

A Multilayer Perceptron (MLP) is a type of Artificial Neural Network (ANN) used for binary classification and prediction problems. It consists of an Input Layer, one or more Hidden Layers, and an Output Layer. Each neuron is connected through weighted links and learns patterns from the training data.

The Backpropagation algorithm is the most commonly used supervised learning algorithm for training an MLP. Its objective is to minimise the prediction error by continuously updating the weights and biases. During training, the network performs a forward pass to generate predictions, calculates the error, and propagates this error backward to improve future predictions. This process continues until the network achieves better accuracy.

2. Forward Pass, Error Calculation and Backpropagation

Backpropagation consists of three major stages:

Step 1: Forward Pass

During the forward pass, the input values are multiplied by their corresponding weights, added with a bias, and passed through an activation function.

The net input is calculated using

$$[z=w_1x_1+w_2x_2+b]$$

where

- x_1, x_2 = Input values
- w_1, w_2 = Weights
- b = Bias

The output is then obtained using the Sigmoid Activation Function.

$$[Output=\frac{1}{1+e^{-z}}]$$

Analytical Example

Parameter	Value
Advertisement Exposure (x_1)	1
Income Level (x_2)	0
Weight (w_1)	0.5
Weight (w_2)	0.3
Bias (b)	0.1
Target Output	1
Forward Pass	

Net Input

$$[z=(1\times 0.5)+(0\times 0.3)+0.1]$$

$$[z=0.6]$$

Predicted Output

$$[\text{Output}=\frac{1}{1+e^{-0.6}}]$$

$$[\text{Output}\approx 0.645]$$

Step 2: Error Calculation

The predicted output is compared with the target output.

The error is calculated using the Mean Squared Error (MSE).

$$[\text{Error}=\frac{1}{2}(\text{Target}-\text{Predicted})^2]$$

$$[\text{Error}=\frac{1}{2}(1-0.645)^2]$$

$$[\text{Error}\approx 0.063]$$

The objective of the algorithm is to minimise this error.

Step 3: Backward Propagation

The calculated error is propagated backward through the network, and the weights are updated using the following equation.

$$[\text{New Weight}=\text{Old Weight}+\eta(\text{Target}-\text{Predicted})\times \text{Input}]$$

Assume the learning rate,

$$[\eta=0.1]$$

Updating Weight w_1

$$[w_1=0.5+0.1(1-0.645)\times 1]$$

$$[w_1=0.5355]$$

Similarly, all remaining weights and biases are updated. These updated values are used in the next iteration, gradually reducing the prediction error and improving the model accuracy.

3. Learning Rate and Activation Functions

The learning rate determines the amount by which the weights are adjusted during each iteration. A suitable learning rate ensures faster convergence and stable training.

Learning Rate Effect

Small	Slow learning but stable convergence
Moderate	Balanced speed and better accuracy
Large	Faster learning but may overshoot the optimum solution

Activation functions introduce non-linearity into the neural network, allowing it to learn complex relationships.

Activation Function Application

Sigmoid	Binary classification
Tanh	Hidden layer processing
ReLU	Faster learning and reduced vanishing gradient
Softmax	Multi-class classification

A proper combination of learning rate and activation function improves the convergence speed and prediction accuracy of the neural network.

4. Advantages, Limitations and Applications

Advantages

- Produces accurate predictions.
- Automatically updates weights.
- Learns complex non-linear patterns.
- Suitable for binary classification.
- Widely used in deep learning models.

Limitations

- Requires large training datasets.
- Training can be computationally expensive.
- Sensitive to learning rate selection.
- May suffer from vanishing gradient problems.

Applications

- Image Classification
- Medical Diagnosis
- Fraud Detection
- Face Recognition
- Email Spam Detection

5. Conclusion

The Backpropagation algorithm is the fundamental learning algorithm used to train a Multilayer Perceptron. It improves the performance of the neural network through three important stages: forward pass, error calculation, and backward propagation. The learning rate controls the speed of weight updates, while activation functions enable the network to learn complex and non-linear relationships. By continuously reducing the prediction error, backpropagation enhances the convergence, accuracy, and overall performance of neural networks, making it one of the most effective algorithms for binary classification problems.

Question 2

In the context of Genetic Algorithms, consider a dataset representing different candidate solutions for maximizing a fitness function based on two parameters: Speed and Accuracy. Each solution is evaluated as either suitable (Yes) or not suitable (No). Using the Genetic Algorithm approach, analyze how the population evolves by applying selection, crossover, and mutation operations. Show the step-by-step improvement of candidate solutions across generations and identify the best optimal solution. Choose your own dataset. Explain how selection chooses the best individuals, how crossover generates new offspring, and how mutation helps in exploring the hypothesis space. Also, provide the final optimized solution after evolution.

1. Introduction :

A **Genetic Algorithm (GA)** is a population-based optimization algorithm inspired by the process of **natural selection and genetics**. It is used to find the best solution to complex optimization and search problems by evolving a population of candidate solutions over several generations.

Each candidate solution is represented as a **chromosome**. During every generation, the algorithm evaluates the fitness of each chromosome, selects the best individuals, performs crossover to create offspring, and applies mutation to introduce diversity. These steps continue until the algorithm finds the optimal or near-optimal solution.

In this problem, the objective is to maximize a fitness function using two parameters: **Speed** and **Accuracy**.

2. Genetic Algorithm and Population Evolution

The Genetic Algorithm begins by creating an initial population of candidate solutions. Each candidate is evaluated using a **fitness function**.

For this example,

Fitness = Speed + Accuracy

The higher the fitness value, the better the candidate solution.

Initial Dataset (Generation 1)

Candidate	Speed	Accuracy	Fitness	Suitable
A	6	7	13	Yes
B	8	6	14	Yes
C	5	9	14	Yes
D	9	8	17	Yes

From the above table, **Candidate D** has the highest fitness value of **17**, making it the best individual in the initial population.

3. Selection, Crossover and Mutation

Selection

Selection is the process of choosing the best individuals based on their fitness values. Candidates with higher fitness have a greater chance of being selected for reproduction.

From the dataset,

- Candidate **D (Fitness = 17)**
- Candidate **B (Fitness = 14)**

are selected as parent chromosomes because they have the highest fitness values.

Crossover

Crossover combines the characteristics of two parent chromosomes to produce new offspring.

Assume the binary representation of the selected parents is

Parent Binary Representation

D 1001

B 1010

Using **single-point crossover**, the last two bits are exchanged.

Parent 1: 10|01

Parent 2: 10|10

After crossover,

- Offspring 1 = **1010**
- Offspring 2 = **1001**

The offspring inherit characteristics from both parents and may produce better solutions.

Mutation

Mutation randomly changes one bit in a chromosome to introduce diversity into the population.

Example:

Before Mutation

1010

After Mutation

1110

Mutation helps the algorithm explore new regions of the search space and prevents the algorithm from getting trapped in local optimum solutions.

4. Evolution of Population

After applying selection, crossover, and mutation, a new generation is produced.

Generation 2

Candidate	Speed	Accuracy	Fitness
Offspring 1	9	8	17
Offspring 2	8	8	16
Mutated Candidate 10	9		19

The mutated chromosome achieves the highest fitness value.

Final Optimized Solution

Parameter Value

Speed 10

Accuracy 9

Fitness 19

The chromosome with **Fitness = 19** is selected as the **optimal solution** after evolution.

5. Advantages, Limitations and Conclusion

Advantages

- Finds optimal solutions efficiently.
- Handles complex optimization problems.
- Explores a large search space.
- Avoids local optimum through mutation.
- Suitable for engineering and machine learning applications.

Limitations

- Computationally expensive for large populations.
- Performance depends on parameter selection.
- May require many generations to converge.
- Mutation rate must be carefully chosen.

Conclusion

The **Genetic Algorithm** is a powerful optimization technique that improves candidate solutions through **selection, crossover, and mutation**. Selection chooses the fittest individuals, crossover combines their characteristics to create new offspring, and mutation introduces diversity to explore new solutions. In this example, the algorithm successfully evolved the population from an initial fitness of **17** to an optimized fitness of **19**. Therefore, Genetic Algorithms provide an efficient approach for solving optimization problems and identifying the best possible solution.

Question 3

In the context of Neural Networks for classification, consider a dataset used to predict whether a patient has a disease based on attributes such as Age, Blood Pressure, Glucose Level, and Lifestyle Risk. The output indicates whether the disease is present (Yes) or not (No). Using this dataset, explain how a neural network model can be trained using backpropagation to classify the data. Also, evaluate the model performance by comparing actual and predicted outputs. Compute the values of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). Based on these values, analyse the performance of the neural network using suitable evaluation metrics. Also, explain how preprocessing and feature selection improve the model's accuracy and reliability.

1. Introduction

Neural Networks are powerful machine learning models used for solving classification problems. In healthcare, they help predict whether a patient has a disease by analysing features such as **Age, Blood Pressure, Glucose Level, and Lifestyle Risk**. During training, the network learns patterns from the dataset using the **Backpropagation algorithm**, which updates the weights to minimise prediction errors.

After training, the model predicts whether a patient has the disease (**Yes**) or not (**No**). The performance of the model is evaluated using a **Confusion Matrix**, which provides values such as **True Positive (TP)**, **True Negative (TN)**, **False Positive (FP)**, and **False Negative (FN)**. These values are further used to calculate evaluation metrics like **Accuracy, Precision, Recall, and F1-Score**.

2. Neural Network Training Using Backpropagation

The neural network learns by processing the training dataset in several steps.

Training Process

1. Input patient attributes into the neural network.
2. Perform a **Forward Pass** to predict the output.
3. Compare the predicted output with the actual output.
4. Calculate the prediction error.
5. Use **Backpropagation** to update the weights.
6. Repeat the process until the error becomes minimum.

The trained model can then classify new patient data accurately.

Sample Dataset

Patient	Age	Blood Pressure	Glucose	Lifestyle Risk	Actual Output	Predicted Output
P1	45	High	High	High	Yes	Yes
P2	38	Normal	High	Medium	Yes	No
P3	29	Normal	Normal	Low	No	No
P4	56	High	High	High	Yes	Yes
P5	34	Low	Normal	Medium	No	Yes

From the above dataset, the neural network predicts disease presence for each patient.

3. Model Performance Evaluation

The predicted outputs are compared with the actual outputs using a **Confusion Matrix**.

Confusion Matrix

Actual / Predicted	Yes	No
Yes	TP = 2	FN = 1
No	FP = 1	TN = 1

Therefore,

- **True Positive (TP) = 2**
- **True Negative (TN) = 1**

- **False Positive (FP) = 1**
- **False Negative (FN) = 1**

Evaluation Metrics

Accuracy

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

$$\text{Accuracy} = \frac{2 + 1}{2 + 1 + 1 + 1}$$

$$\text{Accuracy} = \frac{3}{5} = 60\%$$

Precision

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\text{Precision} = \frac{2}{2 + 1} = \frac{2}{3}$$

$$\text{Precision} = 66.7\%$$

Recall

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{Recall} = \frac{2}{2 + 1} = \frac{2}{3}$$

$$\text{Recall} = 66.7\%$$

F1-Score

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

[
F1=66.7%
]

The obtained values indicate that the neural network correctly classifies most patients but still produces a few incorrect predictions.

4. Preprocessing, Feature Selection, Advantages and Conclusion

Data Preprocessing

Before training the neural network, the dataset must be preprocessed.

- Remove missing or duplicate values.
- Convert categorical data into numerical values.
- Normalize numerical attributes.
- Remove noisy or inconsistent data.

Proper preprocessing improves the quality of the training data and increases prediction accuracy.

Feature Selection

Feature selection identifies the most important attributes that influence disease prediction.

Important features include:

- Age
- Blood Pressure
- Glucose Level
- Lifestyle Risk

Removing irrelevant features reduces computation time, avoids overfitting, and improves model reliability.

Advantages

- Provides accurate disease prediction.
- Learns complex relationships among medical attributes.
- Automatically updates weights using backpropagation.
- Improves decision-making in healthcare.
- Can classify unseen patient data effectively.

Limitations

- Requires sufficient training data.
- Training may take longer for large datasets.
- Sensitive to poor-quality data.
- Performance depends on proper parameter tuning.

5. Conclusion

Neural Networks trained using the **Backpropagation algorithm** are effective for disease classification. The model learns patterns from patient attributes and predicts whether a disease is present or not. Using the confusion matrix, the calculated values are **TP = 2, TN = 1, FP = 1, and FN = 1**, resulting in an **Accuracy of 60%, Precision of 66.7%, Recall of 66.7%, and F1-Score of 66.7%**. Proper **data preprocessing** and **feature selection** improve the model's accuracy, reduce errors, and enhance the overall reliability of the neural network.

